

Browser Testing Scenarios

Manual browser, Swagger UI, and DevTools validation guide for the Spring Boot URL shortener

Purpose

This guide provides browser-oriented testing scenarios for validating the URL shortener after running it locally with Docker Compose or after deploying it to Cloud Run. The tests use a browser, Swagger UI, and browser DevTools without writing code.

Preconditions

- Application is running locally at `http://localhost:8080` or deployed at a Cloud Run URL.
- PostgreSQL is reachable and `/actuator/health` reports UP.
- Redis is reachable for cache-backed redirect testing.
- Use Chrome, Edge, or Firefox with DevTools Network tab open for redirect/header checks.
- Swagger UI is available at `/swagger-ui.html`.

Base URLs

Environment	Base URL	Use when
Local Docker	<code>http://localhost:8080</code>	Running docker compose up --build locally.
Cloud Run	<code>https://YOUR_CLOUD_RUN_URL</code>	Testing deployed service.

Browser Test Matrix

ID	Scenario	Tool	Expected result
BT-01	Health endpoint	Browser address bar	<code>/actuator/health</code> returns status UP.
BT-02	Swagger UI loads	Browser address bar	Swagger page renders APIs.
BT-03	Create short URL	Swagger UI	Returns 201 Created with originalUrl, shortCode, shortUrl, createdAt.
BT-04	Duplicate URL	Swagger UI	Second create returns existing shortCode with 200 OK.
BT-05	Redirect	Browser + DevTools	Short URL returns 302 and opens original URL.
BT-06	Analytics after redirect	Swagger UI	totalClicks increases and recentClicks has browser/device metadata.
BT-07	Invalid URL validation	Swagger UI	Invalid payload returns 400 structured error.
BT-08	Unknown short code	Browser + Swagger UI	Missing code returns 404.
BT-09	Swagger schema review	Swagger UI	Request/response shape matches API contract.
BT-10	Cache behavior smoke check	Repeated browser redirects	Redirect remains consistent and analytics continues increasing.

BT-01 - Health Endpoint

Steps

- Open a browser tab.
- Navigate to `http://localhost:8080/actuator/health` or `https://YOUR_CLOUD_RUN_URL/actuator/health`.
- Observe the JSON response.

Expected Result

```
{"status": "UP"}
```

Failure signals: connection refused, timeout, status DOWN, or a non-JSON error page.

BT-02 - Swagger UI Loads

Steps

- Open `/swagger-ui.html` in the browser.
- Confirm the page title and API operation list are visible.
- Expand POST `/api/v1/urls`, GET `/api/v1/urls/{shortCode}`, and GET `/api/v1/analytics/{shortCode}`.

Expected Result

Swagger UI renders successfully and exposes the URL creation, redirect, and analytics endpoints.

BT-03 - Create Short URL Through Swagger

Steps

- In Swagger UI, expand POST /api/v1/urls.
- Click Try it out.
- Use a valid JSON payload.
- Click Execute and inspect the response.

Expected Result

```
{"originalUrl": "https://www.google.com"}
```

Field	Expected
HTTP status	201 Created for first creation.
originalUrl	Matches submitted URL.
shortCode	Base62-like generated code.
shortUrl	Base URL plus shortCode.
createdAt	ISO timestamp.

BT-04 - Duplicate URL Detection

Steps

- Run the same POST /api/v1/urls request again with the exact same originalUrl.
- Compare the second response with the first response.

Expected Result

The second request returns 200 OK and the same shortCode, proving duplicate detection works.

BT-05 - Redirect Using Browser and DevTools

Steps

- Copy the shortUrl returned from BT-03.
- Open browser DevTools and select the Network tab.
- Paste the shortUrl into the address bar and press Enter.
- Select the redirect request in Network details.

Expected Result

Check	Expected
HTTP status	302 Found for the short URL request.
Location header	Original URL, for example https://www.google.com.
Browser behavior	Browser navigates to the original URL.
Analytics side effect	A click event is persisted.

BT-06 - Analytics After Redirect

Steps

- Return to Swagger UI.
- Expand GET /api/v1/analytics/{shortCode}.
- Click Try it out and enter the shortCode from BT-03.
- Execute the request after at least one redirect.

Expected Result

Field	Expected
shortCode	Matches the tested shortCode.
originalUrl	Matches the original URL.
totalClicks	At least 1 after one browser redirect.

Field	Expected
recentClicks	Contains clickTimestamp, browser, device, and optional referrer.

BT-07 - Invalid URL Validation

Steps

- In Swagger UI, expand POST /api/v1/urls.
- Submit an unsupported or malformed URL.
- Inspect the response body.

Expected Result

```
{"originalUrl":"ftp://example.com/file"}
{"originalUrl":"not-a-url"}
```

The API returns 400 Bad Request with a structured error payload.

BT-08 - Unknown Short Code

Steps

- Open a browser address bar and navigate to /NoSuchCode123.
- In Swagger UI, call GET /api/v1/analytics/NoSuchCode123.

Expected Result

The API returns 404 Not Found for missing redirect and analytics records. The response should not expose a Java stack trace.

BT-09 - Swagger Schema Review

Steps

- Open /swagger-ui.html.
- Expand Schemas or inspect each endpoint request/response section.
- Confirm CreateUrlRequest requires originalUrl.
- Confirm analytics response includes totalClicks and recentClicks.

Expected Result

Swagger documentation matches the implemented API contract and can be used by another developer or tester.

BT-10 - Cache Behavior Smoke Check

Steps

- Use the same shortUrl several times from the browser.
- Keep DevTools Network open.
- After repeated redirects, query analytics again.

Expected Result

Redirect continues to work consistently, and totalClicks increases for each successful redirect. Browser-only testing cannot directly prove Redis hit/miss status, but it validates user-visible cache-path behavior.

Additional Browser Negative Tests

Scenario	Input	Expected result
Blank originalUrl	{"originalUrl":""}	400 Bad Request.
Missing originalUrl	{}	400 Bad Request.
Credential URL	{"originalUrl":"https://user:pass@example.com"}	400 Bad Request.
Unsupported scheme	{"originalUrl":"file:///tmp/a"}	400 Bad Request.
Framework endpoints	Open /swagger-ui.html and /actuator/health	These endpoints should not be intercepted by redirect route.

Evidence to Capture

- Screenshot of /actuator/health returning UP.
- Screenshot of Swagger UI endpoint list.

- Create API response showing shortCode and shortUrl.
- DevTools Network screenshot showing 302 redirect and Location header.
- Analytics response showing totalClicks incremented.
- Error response screenshot for invalid URL.

Troubleshooting Checklist

Symptom	Likely cause	Action
Browser cannot connect	Application is not running or port 8080 is blocked.	Run docker compose ps and check app logs.
Health is DOWN	PostgreSQL or Redis dependency is unavailable.	Check docker compose logs postgres redis app.
Swagger returns 404	App did not start fully or SpringDoc dependency is missing.	Check startup logs and build.gradle.
Create returns 500	Database connection or schema issue.	Check datasource environment variables and PostgreSQL logs.
Redirect does not navigate	Short code missing or invalid Location.	Verify create response and DevTools headers.
Analytics stays zero	Redirect request did not complete through app.	Use shortUrl from API and recheck analytics.

Exit Criteria

- Health endpoint returns UP.
- Swagger UI loads and exposes the required endpoints.
- Valid URL creation returns a shortCode and shortUrl.
- Duplicate URL creation returns the existing shortCode.
- Short URL redirects to the original URL.
- Analytics count increases after redirects.
- Invalid and missing inputs return proper 400 or 404 responses.